PPSITMAJM402 DEEP LEARNING

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UNIT III: CONVOLUTIONAL NEURAL NETWORK

INTRODUCTION TO CNN

- Artificial Intelligence aims to make machines intelligent
- Computer Vision helps machines “see” and interpret images
- CNN is a Deep Learning algorithm mainly used for:
  - Image classification
  - Object detection
  - Face recognition
- CNN automatically learns features from images
- Requires **less manual feature engineering**

Real-Life Example:

Face unlock in mobile phones uses CNN to recognize your face.

WHAT IS A CONVOLUTIONAL NEURAL NETWORK?

- CNN is a special type of neural network
- CNN is a subset of Deep Learning
- Automatically learns features
- Inspired by human brain
- Requires large data and high computation
- Designed to process image and visual data
- Takes image as input and predicts a class label
- Uses layers like:
  - Convolution layer
  - Pooling layer
  - Fully connected layer
  - Softmax output

Example:

Input → Image of car

Output → Car / Truck / Van

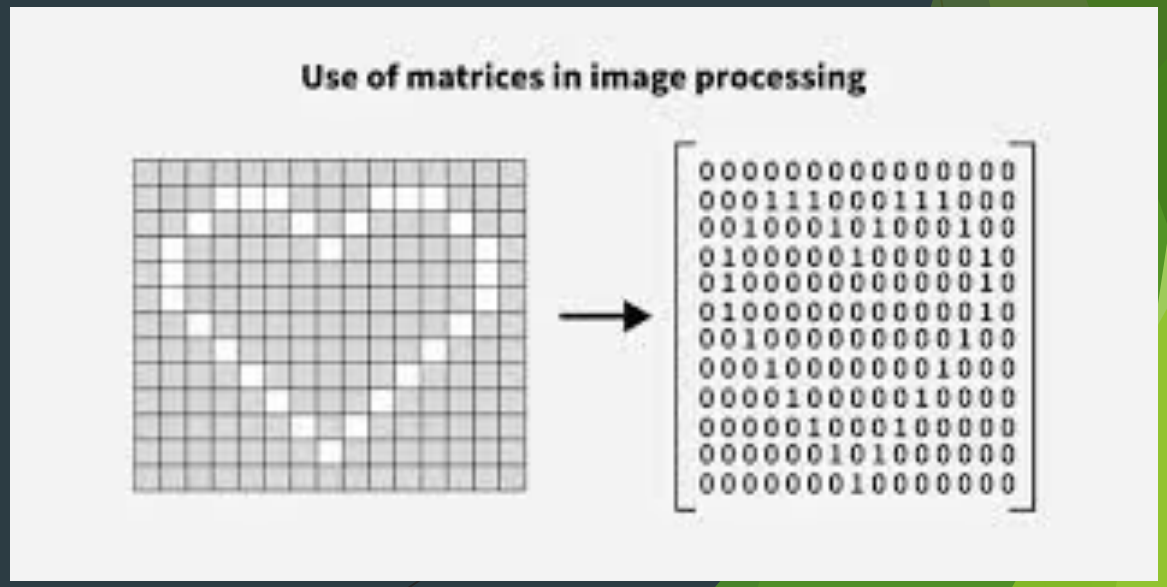
IMAGE AS A MATRIX

- Computer sees images as numbers (pixels)
- Image dimensions: **Height × Width × Depth**
- Grayscale image → Depth = 1
- RGB image → Depth = 3 (Red, Green, Blue)

Example:

6×6×3 → Color image

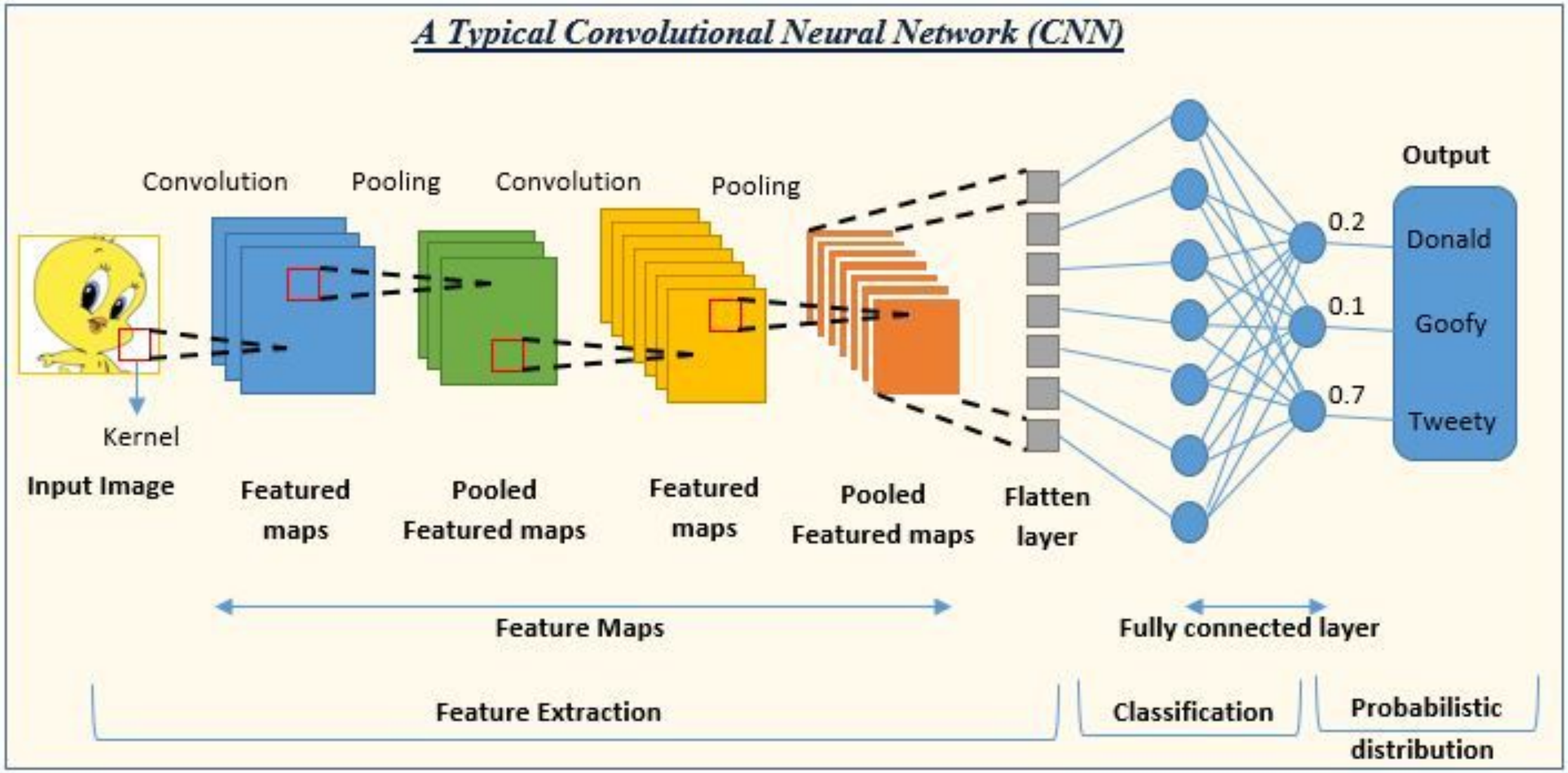
4×4×1 → Grayscale image



BASIC CNN ARCHITECTURE

- Input Image
- Convolution Layer
- Pooling Layer
- Fully Connected Layer
- Softmax Layer (classification)

Real-Life Example:
Just like human eyes → brain → decision


A Typical Convolutional Neural Network (CNN)
Convolution
Pooling
Convolution
Pooling
Output
0.2
0.1
0.7
Donald
Goofy
Tweety
Kernel
Input Image
Featured maps
Pooled Featured maps
Featured maps
Pooled Featured maps
Flatten layer
Feature Maps
Fully connected layer
Feature Extraction
Classification
Probabilistic distribution

CONVOLUTION LAYER

First and most important layer in CNN

Extracts features like:

- Edges
- Corners
- Textures

Uses **filters (kernels)** to scan image

Preserves spatial relationship between pixels

Source layer

5	2	6	8	2	0	1	2
4	3	4	5	1	9	6	3
3	9	2	4	7	7	6	9
1	3	4	6	8	2	2	1
8	4	6	2	3	1	8	8
5	8	9	0	1	0	2	3
9	2	6	6	3	6	2	1
9	8	8	2	6	3	4	5

Convolutional kernel

-1	0	1
2	1	2
1	-2	0

Destination layer

	5						

$$(-1\times5) + (0\times2) + (1\times6) + (2\times4) + (1\times3) + (2\times4) + (1\times3) + (-2\times9) + (0\times2) = 5$$

CONVOLUTION OPERATION

- Mathematical operation between:
  - Image matrix
  - Filter (kernel)
- Filter slides over image
- Produces a **Feature Map**

Example:

Edge detection filter highlights boundaries in an image

FILTER / KERNEL

- Small matrix (usually 3×3 or 5×5)
- Applied repeatedly on image
- Learns patterns automatically during training

Types of Filters:

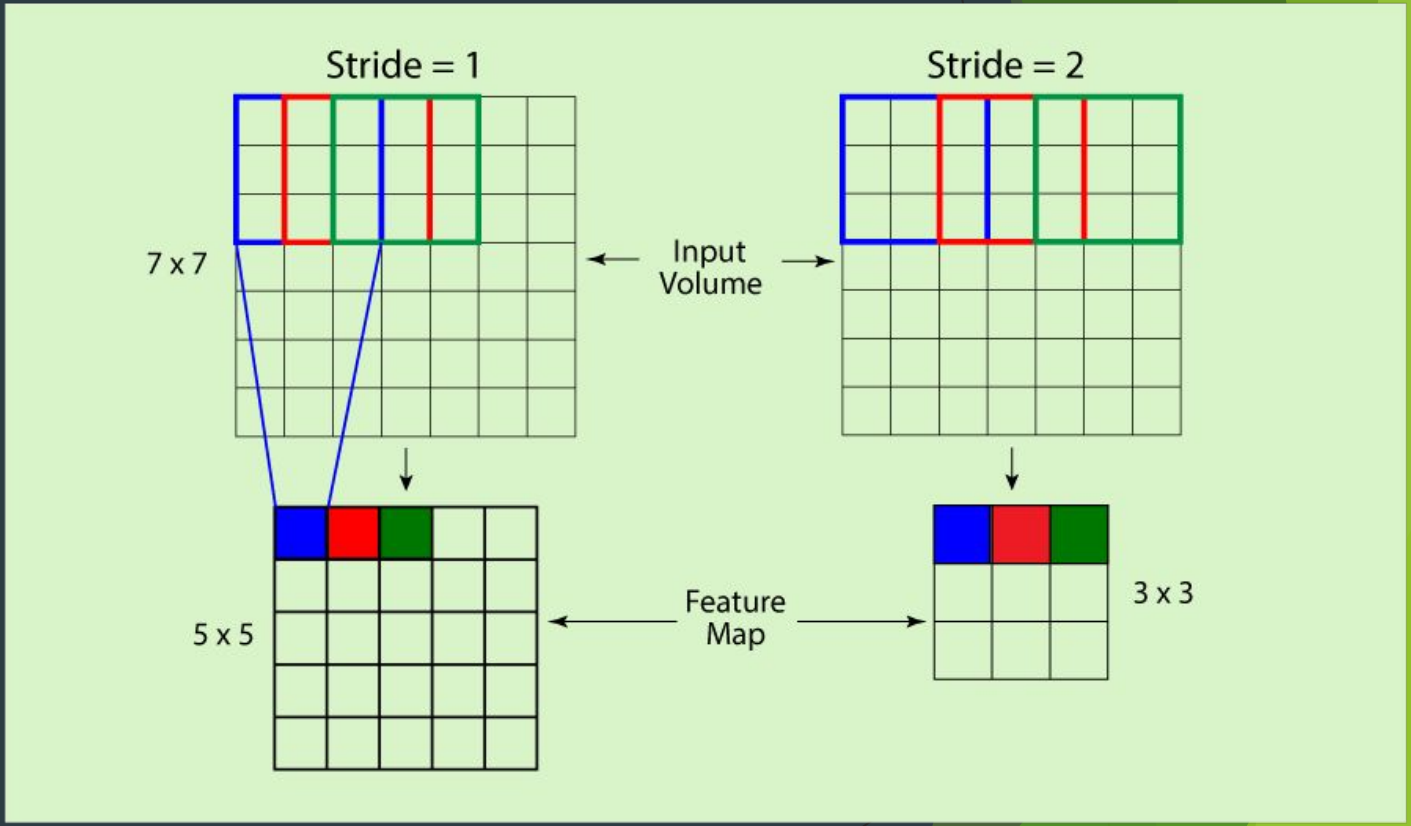
- Blur
- Sharpen
- Edge detection

Real-Life Example:
Photo editing apps use filters for sharpening and smoothing.

STRIDE

- Stride = number of pixels filter moves at a time
- When the stride is equalled to 1, then we move the filters to 1 pixel at a time
- Similarly, if the stride is equalled to 2, then we move the filters to 2 pixels at a time.

Real-Life Example:
Scanning text letter-by-letter vs skipping every second letter.



PADDING

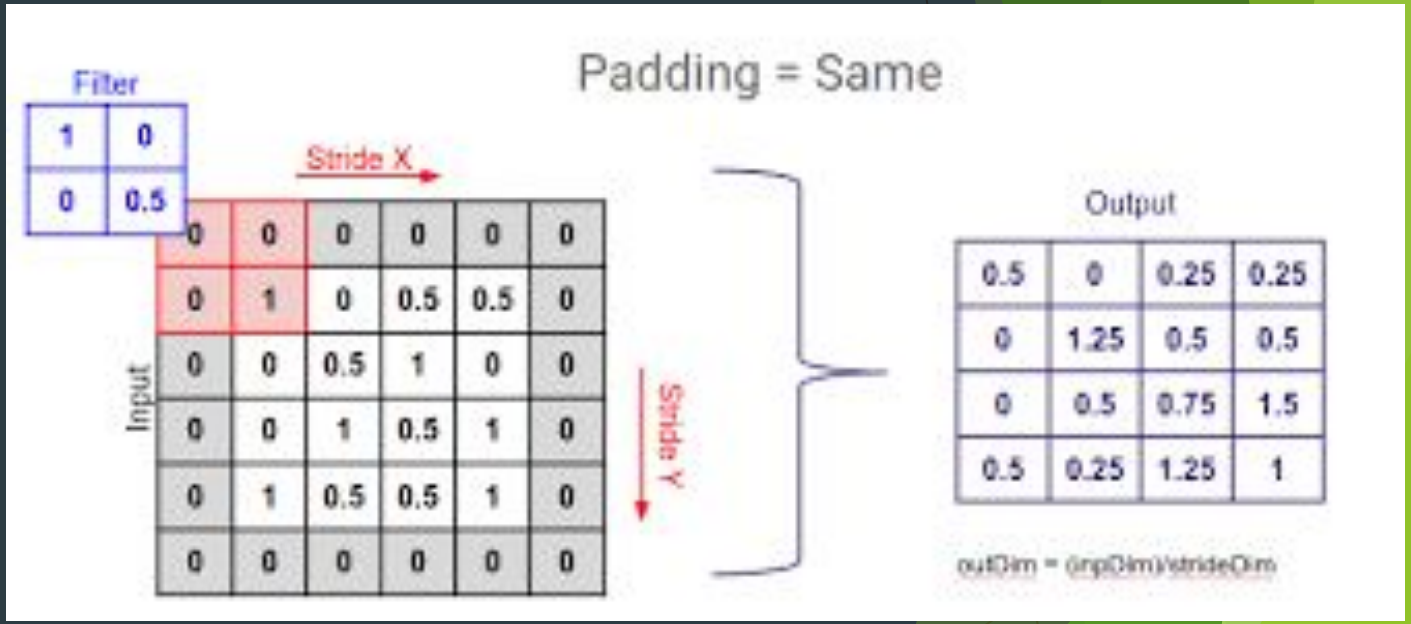
- Padding adds extra pixels (usually zeros) around image
- Prevents image shrinking
- Helps preserve edge information

Problems without Padding:

- Image size reduces
- Loss of corner information

Real-Life Example:

Adding margins to a photo so content is not cut.



WHY CNN OVER FEED-FORWARD NN?

- Feed-Forward NN flattens image → loses spatial info
- CNN:
  - Uses fewer parameters
  - Preserves spatial structure
  - Reuses weights
- Inspired by human visual cortex

Example:

Recognizing a face depends on position of eyes, nose, mouth.

SLIDING WINDOW PROTOCOL

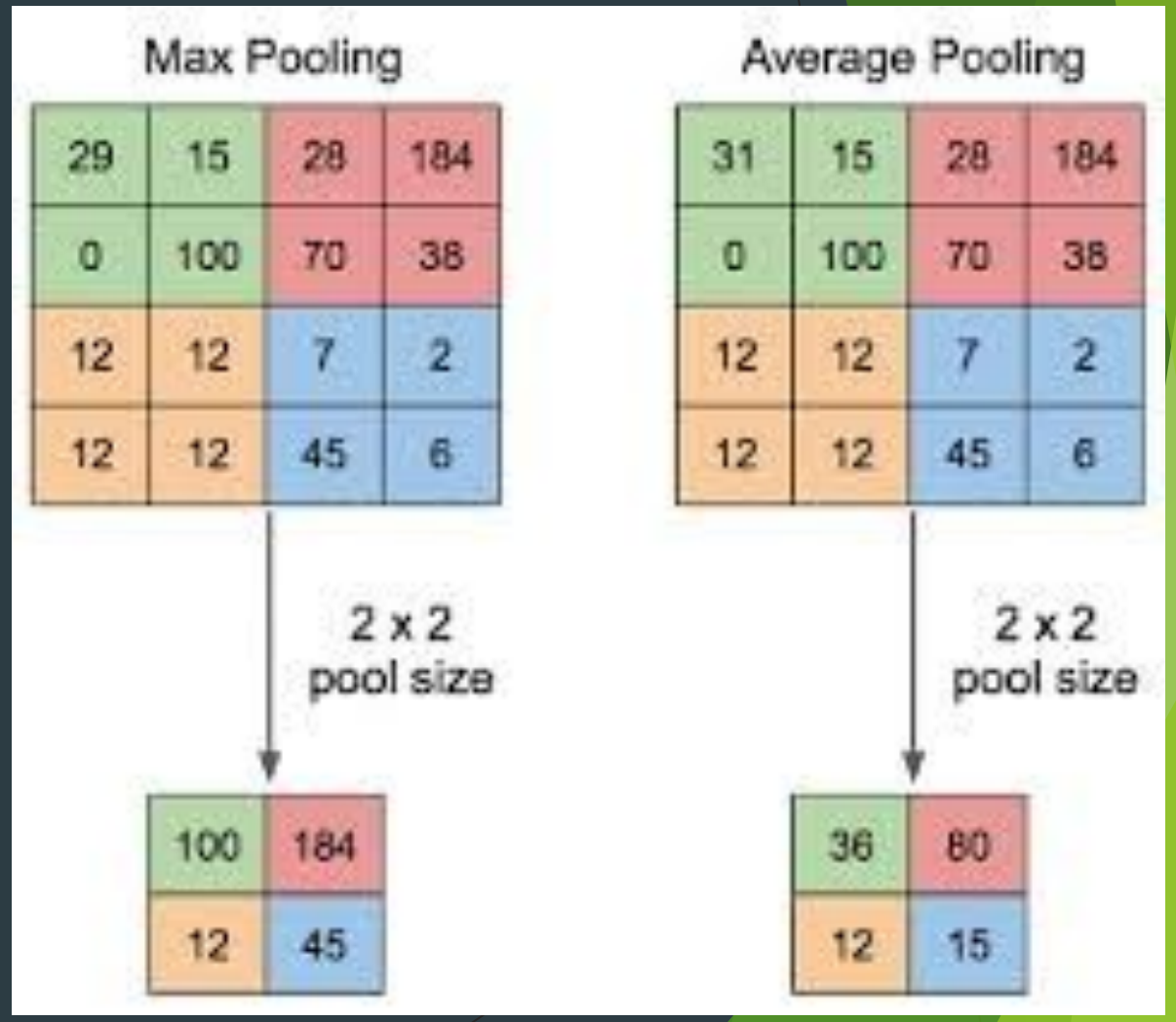
Steps:

1. Kernel starts at top-left corner
2. Performs dot product
3. Moves right
4. Moves down row by row
5. Continues till entire image is covered

POOLING

- Reduces image size
- Reduces number of parameters
- Prevents overfitting
- Retains important information

Analogy:
Summarizing a paragraph instead of reading every word.



TYPES OF POOLING

MAX POOLING

- Takes maximum value from region
- Keeps strongest feature
- Most commonly used pooling

AVERAGE POOLING

- Takes average of values
- Smoothens feature map
- Less commonly used

SUM POOLING

- Adds or averages values
- Similar to average pooling
- Used in specific applications

DATA TYPES IN CNN

CNN supports:

- Single-channel data (1D Data)- Audio waveform, Time-based signals
- Multi-channel data (2D Data) - Grayscale images, Audio spectrograms
- Variable input sizes (3D Data) - Color images (RGB), Medical scans (CT, MRI)
- VIDEO DATA - Time × Height × Width

  Used in: Surveillance, Action recognition

CNN VS RNN

CNN	RNN
Image processing	Text / Speech
Spatial data	Sequential data
Fixed input size	Variable input
No memory	Has memory

CHAPTER 2: SEQUENCE MODELLING

Why Sequence Modelling is Different from FNN & CNN

- In **Feedforward Neural Networks (FNN)** and **CNN**, we assume every input is independent of the previous input.
- But in real life?
  - Words in a sentence depend on previous words
  - Notes in music depend on previous notes
  - Stock price today depends on yesterday
  - Speech depends on previous sound
- **The past matters.** That's where **Sequence Modelling** comes in.
- Sequence modelling = **predicting what comes next based on what came before.**
- Example: "I am going to the ____"
- You already know the next word is likely **market / school / office** because of previous words.
- A normal neural network cannot remember previous words. An RNN **can remember.**

Auto-Completion (Real Life Example)

- When you type in Google: “best laptop for …”
- Google predicts the rest.
- How? It has learned sequences of words from millions of searches.
- This is sequence modelling.

Part of Speech Tagging

- POS tagging = identifying role of each word.
- Example: “Sophia purchased a new laptop from Apple store.”

Word	Tag	Why
Sophia	Noun	person
purchased	Verb	action
new	Adjective	describes laptop
laptop	Noun	object
Apple store	Noun	place

- The model must understand word relationships in a sentence.
- This requires remembering previous words → RNN.

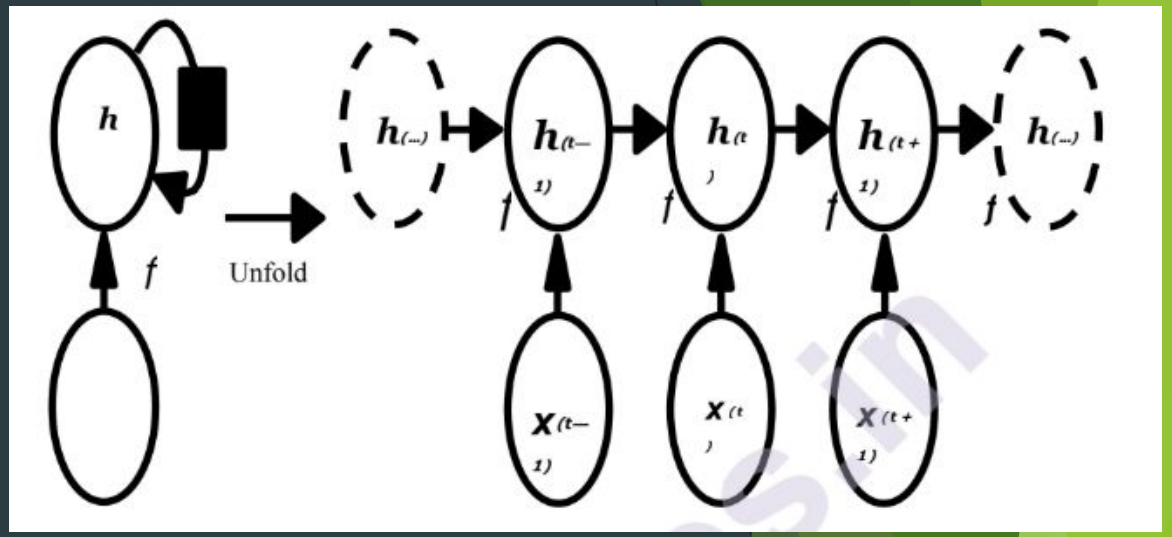
Sequence Classification

Here, we give a sequence → predict a category.

Input	Output
Audio clip	Text transcript (Speech recognition)
Sentence	Sentiment (positive/negative)
DNA sequence	Protein type
French sentence	English sentence
Video frames	Activity

Notice: Input length is not fixed.

Unfolding Computational Graphs



- RNN looks like a loop: h(t) → h(t+1) → h(t+2)
- This loop is hard to understand.
- So we unfold it across time:

  x1 → h1 → y1

  x2 → h2 → y2

  x3 → h3 → y3

- But here's the key: **SAME WEIGHTS** are used at every time step. That's **parameter sharing.**
- **Why important?**
  - Fewer parameters
  - Learns patterns across time
  - Can handle any sequence length

What is happening inside RNN?

At time t:

$$a^t = g(W_{aa}a^{t-1} + W_{ax}x^t + b)$$

$$y^t = g'(W_{ya}a^t + b_y)$$

- Current state = previous memory + current input
- RNN has memory (hidden state).

Problem of Simple RNN

- It only looks backward.
- While predicting a word, sometimes future words help.
- Example:

  "He went to the bank to deposit money"

  vs

  "He sat on the river bank"

- To know meaning of bank, we need future word.
- Solution: **Bi-Directional RNN.**

Types of RNN

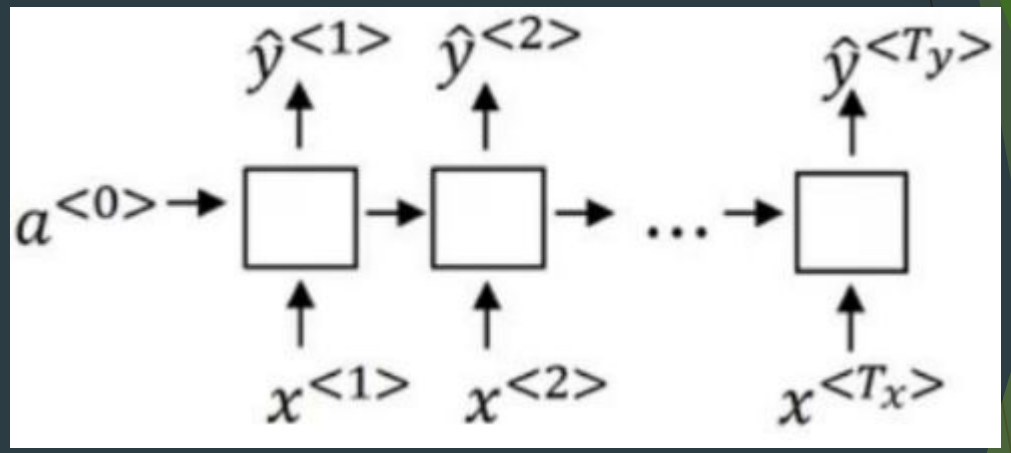
1. Many-to-Many

   Input word → Output word

   Example: Named Entity Recognition

   Sophia | is | from | Mumbai

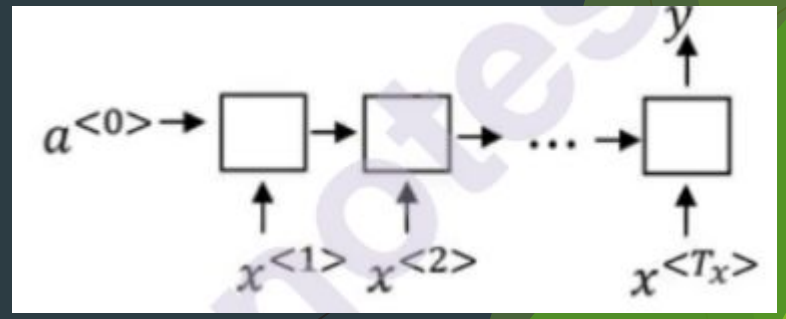
   Noun | O | O | Place



1. Many-to-One

   Sentence → Single output

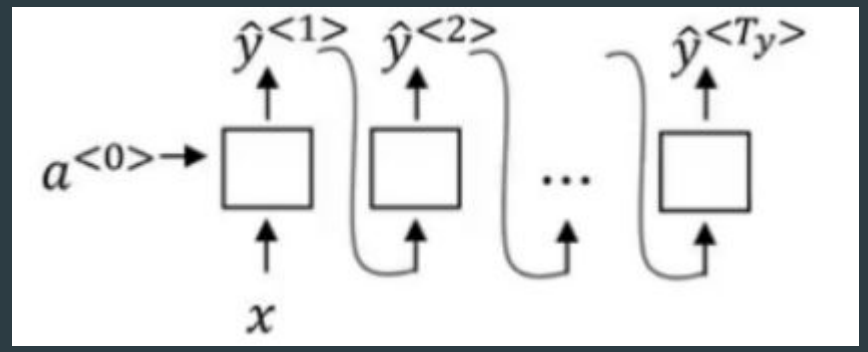
   Example: Sentiment ("This movie is amazing" → Positive)



3. One-to-Many

Single input → Sequence output

Example: Music → Lyrics generation

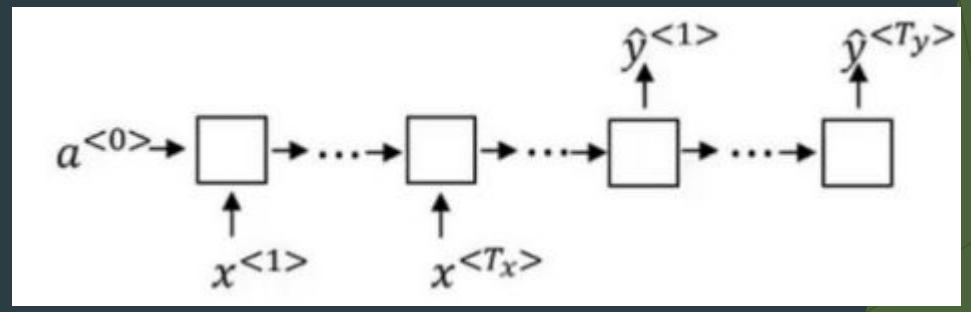


4. Encoder–Decoder (Most Used in Industry)

Many-to-Many but unequal length

Example: Machine Translation

French sentence → English sentence



Encoder reads full sentence → Decoder generates output.

Word Embeddings

- One hot vector:

  Apple = [0 0 0 1 0 0]

  Orange = [0 0 1 0 0 0]

- Dot product = 0
- Model thinks Apple and Orange are unrelated.
- But in reality, they are both fruits.
- **Word Embedding solves this:**

  Apple = [0.21, 0.87, 0.45]

  Orange = [0.25, 0.90, 0.40]

  King = [0.99, 0.10, 0.88]

- Now Apple and Orange are close.

Word2Vec

- Goal: Predict a word using context.
- Sentence: “I want a glass of orange juice”
- Predict: juice
- Different contexts:
  - Last 4 words
  - Left & right words
  - Only 1 nearby word (Skip-Gram)
- All help in learning meaningful embeddings.

Applications of Word Embeddings

Sentiment Classification (Simple model)

Take average of word vectors → classify.

Problem: ignores word order. “Not good” and “good” look similar.

Better model: Use RNN with embeddings.

Debiasing Word Embeddings

- Model learns bad stereotypes: Man : Programmer :: Woman : Homemaker
- Steps to fix:
  - Find bias direction using pairs:
  - she-he
  - girl-boy
  - mother-father
- Create gender vector g
- Neutralize words like:
  - computer
  - singer
  - doctor
- By removing projection on g.

CHAPTER 3: APPLICATIONS

INTRODUCTION

- Deep learning is inspired by how the human brain processes patterns.
- Just like humans recognize faces, understand speech, and interpret language, deep learning models learn patterns from huge data and perform similar intelligent tasks.
- The power of deep learning lies in its ability to automatically discover patterns from raw data.
- Deep learning models become more powerful when:
  - Data increases
  - Model size increases
  - Bigger data + bigger models = better accuracy
  - Computation power increases
- But large models cannot be trained on a single machine. Hence, distributed training using GPUs/TPUs is used.
- Challenges: communication cost, memory limits, accuracy drop with large batch

Fast CPU & GPU & Distributed Implementations

- Initially, neural networks were trained only on CPUs.
- Researchers optimized mathematical operations to squeeze maximum performance from CPUs before GPUs became popular.
- GPUs were made for video games but turned out perfect for deep learning because they can perform thousands of parallel operations, ideal for matrix multiplication in neural networks.
- Faster than CPUs for DL tasks
- Two methods:
  - Data Parallelism: Same model on many machines, different data
  - Model Parallelism: Model split across machines
- Used when models have billions of parameters.

Aspect	Fast CPU Implementations	GPU Implementations	Distributed Implementations
Initial Usage	Early neural networks were trained only on CPUs	Introduced later when DL computation increased	Needed when models/data became too large for one machine
Main Idea	Optimize mathematical operations on a single processor	Use massive parallel processing power	Use many machines together to train one model
Hardware Used	Single machine CPU	Graphics Processing Unit (GPU)	Cluster of CPUs/GPUs/TPUs
Computation Style	Sequential processing	Parallel processing (thousands of cores)	Parallel + distributed across machines
Speed	Slow for large neural networks	Much faster than CPU for DL tasks	Fastest for very large-scale models

Memory Capacity	Limited to one machine's RAM	Higher memory than CPU	Combined memory of multiple machines
Optimization Focus	Numerical routines, floating-point optimization	Matrix multiplication, tensor operations	Communication efficiency, workload distribution
Scalability	Very limited	Moderate	Very high (billions/trillions of parameters)
Parallelism Type	Minimal	Data parallel within GPU cores	Data Parallelism & Model Parallelism
Major Challenge	High computation time	Hardware cost	Communication overhead between machines
Best Suited For	Small models, research experiments	Medium to large DL models	Extremely large datasets and models
Real-world Use	Early ML/DL research	Image processing, NLP training, CNNs, RNNs	Google, OpenAI, Meta training massive models
Example Scenario	Training a small ANN on laptop	Training CNN for image classification	Training ChatGPT / large language models

Computer Vision

- Computer vision allows machines to see and understand images like humans.
- CNNs are mainly used.
- Tasks include object recognition, face detection, medical imaging, and autonomous driving.
- Preprocessing: Images must be standardized (same size, pixel range). Some models accept variable sizes, but usually resizing is done.
  - Normalize pixels [0,1] or [-1,1]
  - Resize images
  - Cropping/scaling
- Dataset Augmentation: Instead of collecting more images, we modify existing ones: Rotate, Flip, Zoom, Crop
- This improves model accuracy.

Speech Recognition

- Speech recognition converts audio into text.
- The audio is broken into frames (20ms).
- RNNs/Transformers learn mapping from sound sequence to word sequence.
- Example: Alexa, Google Assistant.

Natural Language Processing

- NLP enables computers to understand human language.
- Used in translation, chatbots, sentiment analysis, summarization.
- n-grams: n-grams predict next word using previous (n−1) words.
- Example (bi-gram): “deep learning” occurs frequently → high probability.
- Simple counting method.
- Hierarchical Softmax: When vocabulary is huge (10,000+ words), calculating softmax is costly. Hierarchical softmax reduces computation from |V| to log|V| using a tree.

Other Applications

- Recommender Systems
  - Used by Amazon, Netflix, Flipkart. Predicts what a user will like.
  - Based on user-item interaction.
- Exploration vs Exploitation
  - System must choose:
    - Recommend what user already likes (exploit)
    - Try new items to learn preference (explore)
  - aThis is contextual bandit problem.
- Knowledge Representation & QA
  - Word embeddings store semantic knowledge. Combining this with reasoning leads to Question Answering systems.
  - Example: ChatGPT, Google search answers.